

We are going to prove that a measure with the maximal entropy cannot be carried by a subset of $S_\beta \setminus P_\beta^\mathbb{Z}$.

We define on the tower (Ω, T) of P_β the map f by:

$$f((x_n)_{n \in \mathbb{Z}}, i) = (y_n)_{n \in \mathbb{Z}} \quad (70)$$

where $x_k \in \mathfrak{C}_\beta$, $y_i \in \{0, 1, \dots, d_1\}$, y_0 denotes the i -th letter of x_0 .

Example 4 Let β such that $d(l_\beta, \beta) = \bullet 2(1)^\infty$. We set $D = \{2(11)^k, k \in \mathbb{N}\}$

$$\mathfrak{C}_\beta = \{0, 1, x2(11)^k 10 | x \in D^* \cup \{\varepsilon\}\}. \quad (71)$$

We set $x_n = 210$ for $n \leq -1$, $x_0 = 21110$ and $x_n = 2210$. Then

$$f((x_n)_{n \in \mathbb{Z}}, 2) = \dots 210 \ 210 \ 210 \ 21 \bullet 110 \ \overline{2210}.$$

Remark 6 It is easy to check that $f \circ T = \sigma \circ f$ (see [14]) and f is one to one. Let $(z_n)_{n \in \mathbb{Z}} \in P_\beta^\mathbb{Z}$, $(x_n)_{n \in \mathbb{Z}}$ such that $x_n \in P_\beta$ and

$$\dots x_{-m} \dots x_{-1} x_0 x_1 x_2 \dots x_m \dots = \dots z_{-n} \dots z_{-1} z_0 \bullet z_1 \dots z_n \dots$$

where z_0 is the i -th letter of x_0 . The inverse map f^{-1} of f from $P_\beta^\mathbb{Z}$ to Ω is given by

$$f^{-1}((z_n)_{n \in \mathbb{Z}}) = ((x_n)_{n \in \mathbb{Z}}, i). \quad (72)$$

The map f is one to one and $f \circ T = \sigma \circ f$. Each σ -invariant measure π on $P_\beta^\mathbb{Z}$ generates a measure $\pi \circ f$ on Ω with the same entropy. Thus the uniqueness of the measure of the entropy $\log |\beta|$ on Ω implies that there is a unique measure with the same entropy on $P_\beta^\mathbb{Z}$.

4.2.1 Hybrid admissible sequences

Definition 11 In what follows, an admissible sequence $(x_i)_{i \in \mathbb{Z}}$ shall be said hybrid if there is an integer $k \in \mathbb{Z}$ such that one of the following conditions hold:

- (a) $(x_i)_{i \geq k} \in \mathfrak{C}_\beta^\mathbb{N}$ and $(x_i)_{i < k} \notin \mathfrak{C}_\beta^\mathbb{N}$;
- (b) $(x_i)_{i \geq k} \notin \mathfrak{C}_\beta^\mathbb{N}$ and $(x_i)_{i < k} \in \mathfrak{C}_\beta^\mathbb{N}$;
- (c) $(x_i)_{i \geq k} \in \Delta_{j,\beta}^\mathbb{N}$ and $(x_i)_{i < k} \notin \Delta_{j,\beta}^\mathbb{N}$;
- (d) $(x_i)_{i \geq k} \notin \Delta_{j,\beta}^\mathbb{N}$ and $(x_i)_{i < k} \in \Delta_{j,\beta}^\mathbb{N}$.

In what follows, we denote by N_β , the set of hybrid sequences.

Example 5 • Let k such that $d_{2k+1} \geq 1$ and $d_1 \dots d_{2k} 0$ be admissible. Then $d_1 \dots d_{2k} 0 \in \mathfrak{C}_\beta$. The sequence $\dots d_1 d_1 d_1 \dots d_1 \bullet (d_1 \dots d_{2k} 0)^\infty$ (infinite repetition of d_1 at left and infinite repetition of $d_1 \dots d_{2k} 0$ at right) is an admissible hybrid sequence.

- We suppose that $d_1 d_2 \dots = 2(1)^\infty$. One has $210 \in \mathfrak{C}_\beta$. The following sequences are hybrid: $\dots 210 210 210 \dots 210 \bullet (2)^\infty$, $\dots 2222 \dots 2 \bullet (210)^\infty$, $\dots 210 210 \dots 210 \bullet 2(1)^\infty$, $\dots 211 211 \dots 211 \bullet 2(1)^\infty$.

Let $N_{\beta,k}$ be the subset of N_β of sequences $(x_i)_{i \geq \mathbb{Z}}$ such that one of the following situations occurs:

- (e) $(x_i)_{i \leq k} \notin \mathfrak{C}_\beta^\mathbb{N}$, there is m such that $x_{k+1} \dots x_{k+m-1} = d_1 \dots d_{m-1}$ and $(-1)^m(d_m - x_{k+m}) < 0$, $x_{k+m} \neq d_1$ and $(x_i)_{i \geq k+1} \in \mathfrak{C}_\beta^\mathbb{N}$,
- (f) $(x_i)_{i \leq k} \in \mathfrak{C}_\beta^\mathbb{N}$ and $(x_i)_{i \geq k+1} \notin \mathfrak{C}_\beta^\mathbb{N}$;